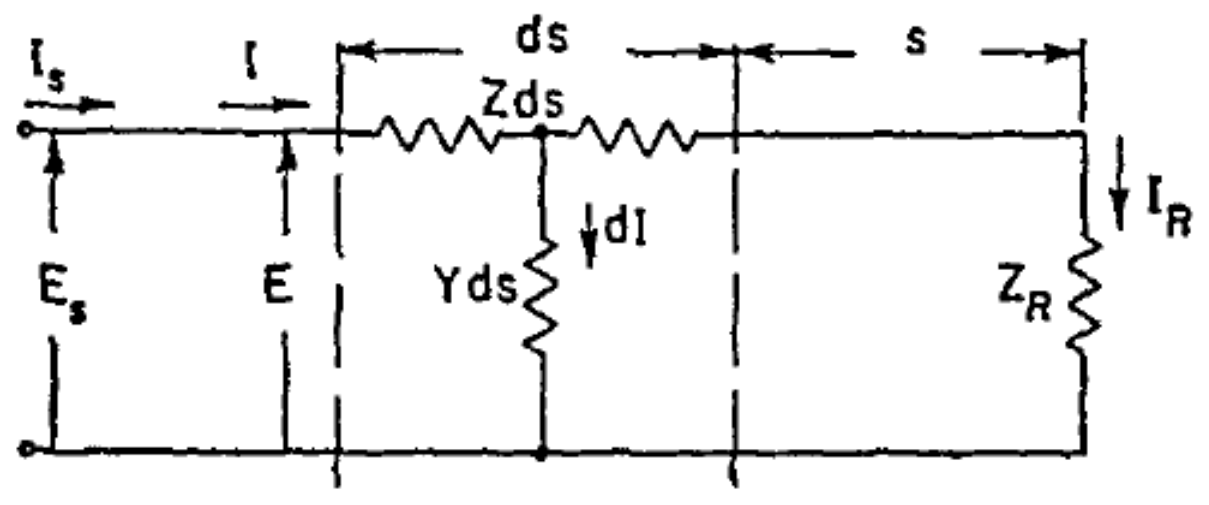
Transmission Line Equation

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Transmission Line Equation – General solution



R → Series resistance (Ω/unit length)

L → Series inductance (H/unit length)

C → Shunt capacitance (F/unit length)

G → Shunt conductance (℧/unit length)

$Z=R+j\omega L$

$Y=G+j\omega C$

Z_R → Receiving end impedance

E_R → Receiving end voltage

I_R → Receiving end current

$Z_R = E_R / I_R$

S → Distance to the point of observation from receiving end

- Consider a TL made up of cascaded structure of infinite no. of T sections
- Let ds be the elemental length of one T section
- There will be a voltage drop dE because of series impedance Zds

$$dE = I\,Zds$$

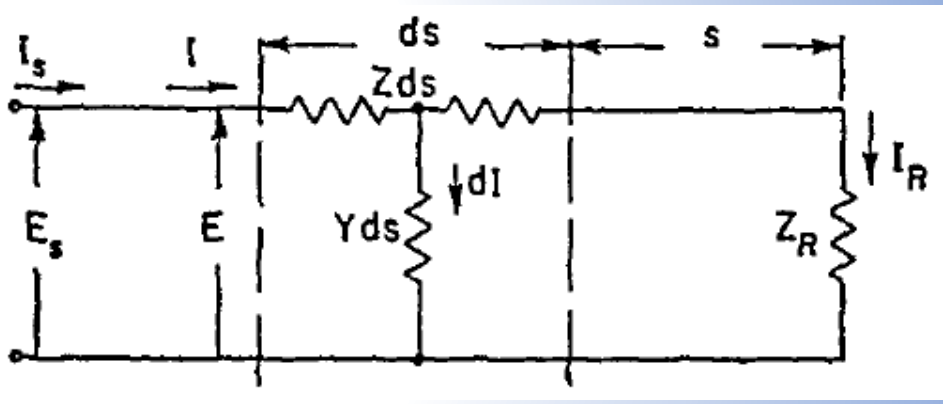
$$\frac{dE}{dS} = IZ \quad \text{---(1)}$$

- There will be a small current dI because of shunt admittance Yds

$$dI = E\,Yds$$

$$\frac{dI}{dS} = EY \quad \text{---(2)}$$

- Differentiate (1) and (2) w.r.t. s



- $\frac{dE}{dS}$ = I Z ---(1)
- $\frac{dI}{dS}$ = E Y ---(2)
- Differentiate (1) and (2) w.r.t. s
- Gives fundamental differential line equations

$\frac{d^2E}{ds^2}$ = Z $\frac{dI}{ds}$ $= Z\ E\ Y$ ----(3)

$\frac{d^2I}{ds^2}$ = Y $\frac{dE}{ds}$ $= Y\ I\ Z$ ----(4)

The solution of these second order differential equation can be used to find voltage and current at any point on the TL

Replace $\frac{d^2}{ds^2}$ terms in Eqn. (3) and (4) with D^2 we get

D^2 E = Z E Y → D = $\pm\sqrt{ZY}$

D^2 I = Y I Z → D = $\pm\sqrt{ZY}$

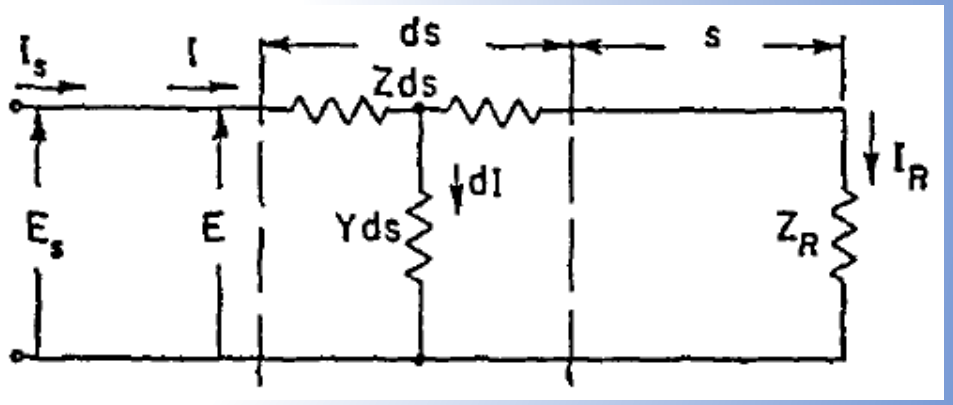
- The solution can be written using these parameters as

$E = A\, e^{\sqrt{zy}\, s} + B\, e^{-\sqrt{zy}\, s}$ ---(5)

$I = C\, e^{\sqrt{zy}\, s} + D\, e^{-\sqrt{zy}\, s}$ ---(6)

where A,B,C and D are constants

The values of these constants are computed by applying the boundary conditions.



At s = 0, $E = E_R$ and $I = I_R$

$E_R = A + B$ ---(7)

$I_R = C + D$ ---(8)

Differentiating (5) and (6) w.r.t. ‘s’ we get

$\frac{dE}{dS} = A\sqrt{ZY}\, e^{\sqrt{zy}\, s} - B\sqrt{ZY}\, e^{-\sqrt{zy}\, s}$ ---(9)

$\frac{dI}{dS} = C\sqrt{ZY}\, e^{\sqrt{zy}\, s} - D\sqrt{ZY}\, e^{-\sqrt{zy}\, s}$ ---(10)

- Substitute the value of (1) and (2) in eqn.s (9) and (10) respectively

$\frac{dE}{dS}$ = I Z ---(1)

$\frac{dI}{dS}$ = E Y ---(2)

$$\frac{dE}{dS} = A\sqrt{ZY}e^{\sqrt{zy}\,s} - B\sqrt{ZY}e^{-\sqrt{zy}\,s} \quad \text{---(9)}$$

$$\frac{dI}{dS} = C\sqrt{ZY}e^{\sqrt{zy}\,s} - D\sqrt{ZY}e^{-\sqrt{zy}\,s} \quad \text{---(10)}$$

$$I\,Z = A\sqrt{ZY}e^{\sqrt{zy}\,s} - B\sqrt{ZY}e^{-\sqrt{zy}\,s}$$

$$E\,Y = C\sqrt{ZY}e^{\sqrt{zy}\,s} - D\sqrt{ZY}e^{-\sqrt{zy}\,s}$$

$$I = A\sqrt{Y/Z}e^{\sqrt{zy}\,s} - B\sqrt{Y/Z}e^{-\sqrt{zy}\,s} \quad \text{---(11)}$$

$$E = C\sqrt{Z/Y}e^{\sqrt{zy}\,s} - D\sqrt{Z/Y}e^{-\sqrt{zy}\,s} \quad \text{---(12)}$$

Applying boundary condition, At s = 0, E = E_R and I = I_R

$$I_R = A\sqrt{Y/Z} - B\sqrt{Y/Z} \qquad E_R = C\sqrt{Z/Y} - D\sqrt{Z/Y}$$

$I_R = A\sqrt{Y/Z} - B\sqrt{Y/Z}$ $E_R = C\sqrt{Z/Y} - D\sqrt{Z/Y}$

$\rightarrow I_R = \sqrt{Y/Z}$ (A - B)

$\rightarrow E_R = \sqrt{Z/Y}$ (C - D)

$\rightarrow A - B = I_R \sqrt{Z/Y}$ --(13)

$\rightarrow C - D = E_R \sqrt{Y/Z}$ --(14)

$E_R = A + B$ ---(7)

$I_R = C + D$ ---(8)

(7) + (13) gives

$E_R + I_R \sqrt{Z/Y} = 2A$

$\rightarrow$ **$A = E_R/2 + \frac{I_R}{2}\sqrt{Z/Y}$**

(7) – (13) gives

$B = E_R/2 - \frac{I_R}{2}\sqrt{Z/Y}$

Solving (8) and (14) gives

$C = I_R/2 + \frac{E_R}{2}\sqrt{Y/Z}$

$D = I_R/2 - \frac{E_R}{2}\sqrt{Y/Z}$

- Substitute the values of these constants in (5) and (6)

$$E = \left(\frac{E_R}{2} + \frac{I_R}{2}\sqrt{Z/Y}\right) e^{\sqrt{zy}\,s} + \left(\frac{E_R}{2} - \frac{I_R}{2}\sqrt{Z/Y}\right) e^{-\sqrt{zy}\,s}$$

$$I = \left(\frac{I_R}{2} + \frac{E_R}{2}\sqrt{Y/Z}\right) e^{\sqrt{zy}\,s} + \left(\frac{I_R}{2} - \frac{E_R}{2}\sqrt{Y/Z}\right) e^{-\sqrt{zy}\,s}$$

But, $E_R/I_R = Z_R$, $\sqrt{\frac{Z}{Y}} = Z_0$ and $\sqrt{ZY} = \gamma$

$$E = \frac{E_R}{2}\left(1 + \frac{1}{Z_R}Z_0\right) e^{\gamma s} + \frac{E_R}{2}\left(1 - \frac{1}{Z_R}Z_0\right) e^{-\gamma s}$$

$$= \frac{E_R}{2}\left(\frac{Z_R+Z_0}{Z_R}\right) e^{\gamma s} + \frac{E_R}{2}\left(\frac{Z_R-Z_0}{Z_R}\right) e^{-\gamma s} \quad \text{---(15)}$$

$$\mathbf{E = \frac{E_R}{2}\left(\frac{Z_R+Z_0}{Z_R}\right)\left[e^{\gamma s} + \frac{Z_R-Z_0}{Z_R+Z_0}\, e^{-\gamma s}\right]} \quad \mathbf{\text{---(16)}}$$

Similarly we can prove that $\mathbf{I = \frac{I_R}{2}\left(\frac{Z_R+Z_0}{Z_0}\right)\left[e^{\gamma s} - \frac{Z_R-Z_0}{Z_R+Z_0}\, e^{-\gamma s}\right]}$ **--(17)**

$$E = \frac{E_R}{2}\left(\frac{Z_R+Z_0}{Z_R}\right)\left[e^{\gamma s} + \frac{Z_R-Z_0}{Z_R+Z_0}\, e^{-\gamma s}\right] \quad \text{---(16)}$$

$$E = \frac{E_R}{2}\left(\frac{Z_R+Z_0}{Z_R}\right) e^{\gamma s} + \frac{E_R}{2}\left(\frac{Z_R-Z_0}{Z_R}\right) e^{-\gamma s} \quad \text{---(15)}$$

$$I = \frac{I_R}{2}\left(\frac{Z_R+Z_0}{Z_0}\right)\left[e^{\gamma s} - \frac{Z_R-Z_0}{Z_R+Z_0}\, e^{-\gamma s}\right] \quad \text{--(17)}$$

- These are standard expressions for voltage and current at any point on a TL in exponential form

$$\mathbf{E = \frac{E_R}{2}\left(\frac{Z_R+Z_0}{Z_R}\right)\left[e^{\gamma s} + K\, e^{-\gamma s}\right]}$$

$$\mathbf{I = \frac{I_R}{2}\left(\frac{Z_R+Z_0}{Z_0}\right)\left[e^{\gamma s} - K e^{-\gamma s}\right]}$$

- Here K = $\frac{Z_R-Z_0}{Z_R+Z_0}$, the reflection coefficient = ratio of amplitudes of the reflected and incident voltage or current wave at the receiving end of the line

K = $\frac{|E_R|}{|E_i|}$ at s = 0 From Eqn. (15)

$$= \frac{\frac{E_R}{2}\left(\frac{Z_R - Z_0}{Z_R}\right)}{\frac{E_R}{2}\left(\frac{Z_R + Z_0}{Z_R}\right)} = \frac{Z_R-Z_0}{Z_R+Z_0}$$

When the line is terminated in its characteristic impedance, $Z_R = Z_0 \rightarrow K = 0$

$$E = \overbrace{\frac{E_R}{2}\left(\frac{Z_R+Z_0}{Z_R}\right)e^{\gamma s}}^{E_i} + \overbrace{\frac{E_R}{2}\left(\frac{Z_R-Z_0}{Z_R}\right)e^{-\gamma s}}^{E_R} \quad \text{---(15)}$$

$$= \frac{E_R}{2Z_R}\{Z_R e^{\gamma s} + Z_o e^{\gamma s} + Z_R e^{-\gamma s} - Z_o e^{-\gamma s}\}$$

$$= \frac{E_R}{Z_R}\{Z_R \frac{[e^{\gamma s} + e^{-\gamma s}]}{2} + Z_o \frac{[e^{\gamma s} - e^{-\gamma s}]}{2}\}$$

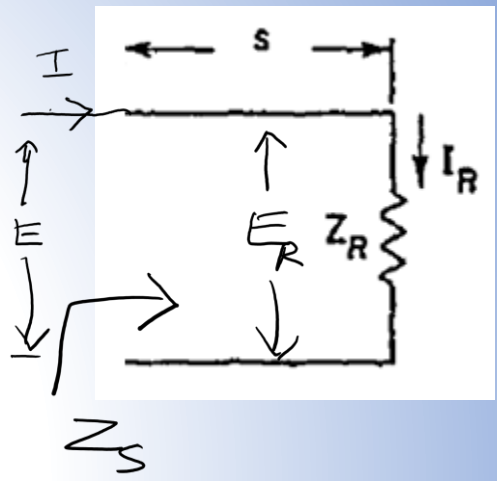
For RF line with $Z_0 = R_0$ and $\gamma = j\beta$ (as $\alpha = 0$)

$$= \frac{E_R}{Z_R}\{Z_R \frac{[e^{j\beta s} + e^{-j\beta s}]}{2} + R_o \frac{[e^{j\beta s} - e^{-j\beta s}]}{2}\}$$

$$E = E_R \cos \beta s + j\, I_R R_0 \sin \beta s$$

Similarly $I = I_R \cos \beta s + j \frac{E_R}{R_0} \sin \beta s$

Input impedance (Z_s):



$$Z_s = \frac{E}{I} = \frac{E_R \cos \beta s + j \frac{E_R}{Z_R} R_0 \sin \beta s}{\frac{E_R}{Z_R} \cos \beta s + j \frac{E_R}{R_0} \sin \beta s}$$

$$= R_0 \frac{Z_R \cos \beta s + j R_0 \sin \beta s}{R_0 \cos \beta s + j Z_R \sin \beta s}$$

$$= R_0 \frac{Z_R + j R_0 \tan \beta s}{R_0 + j Z_R \tan \beta s}$$

Put $Z_R = 0$, Short ckt impedance $Z_{SC} = j R_0 \tan \beta s$

Put Z_R = infinity, Open ckt impedance $Z_{OC} = \lim_{Z_R \to \infty} R_0 \frac{Z_R + j R_0 \tan \beta s}{R_0 + j Z_R \tan \beta s} = \frac{R_0}{j \tan \beta s}$

- $Z_{SC} = j R_0 \tan \beta s$
- $Z_{OC} = \frac{R_0}{j tan \beta s}$
- $Z_{SC} . Z_{OC} = R_0^2$
- $R_0 = \sqrt{Z_{SC} Z_{OC}}$

Voltage and current on the dissipation less line:

- The voltage at any point, distance ‘s’ units from the receiving end of the TL is $\mathbf{E} = \frac{E_R}{2}\left(\frac{Z_R+Z_0}{Z_R}\right)\left[e^{\gamma s} + \mathbf{K}\, e^{-\gamma s}\right]$
- For the line with zero dissipation $\alpha = 0$ and $Z_0 = R_0$ (RF line)

$$E = \frac{E_R}{2}\left(\frac{Z_R+R_0}{Z_R}\right)\left[e^{j\beta s} + K\, e^{-j\beta s}\right]$$

Incident wave $E_i = E_0\, e^{j\beta s}.\, e^{j\omega t}$

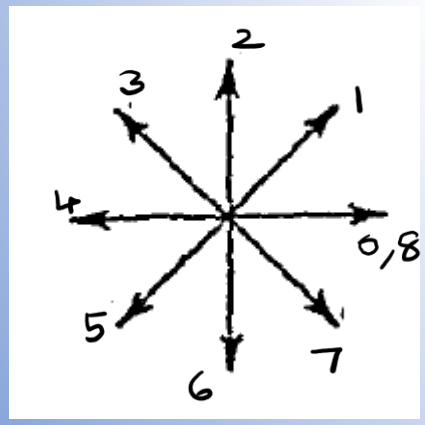
$= E_0 \angle \beta s + \omega t$

- $E_i = E_0 \angle \beta s + \omega t$
- At $t = 0$, $E_i = E_0 \angle \beta s$
- At $s = 0$, $E_i = E_0 \angle 0^\circ$

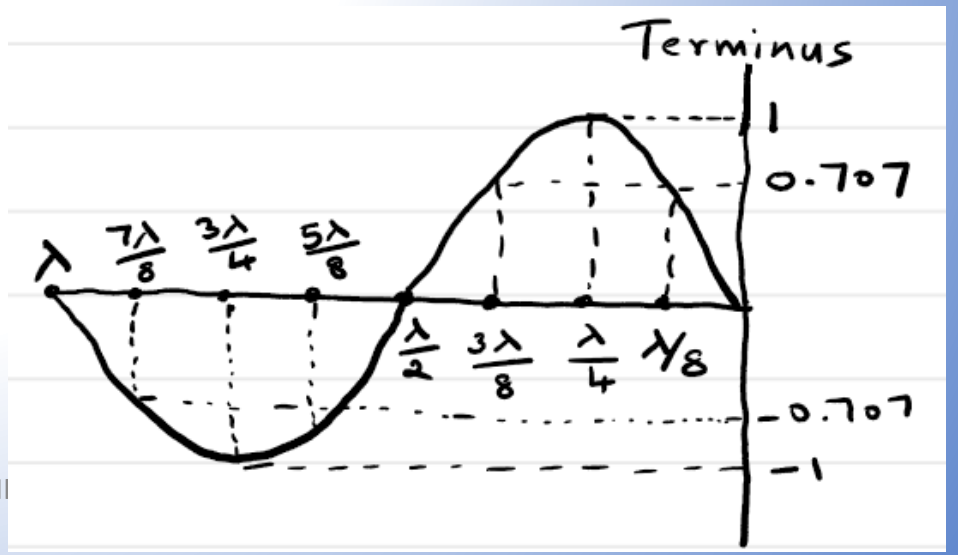
$s = \frac{\lambda}{8}$; $\beta s = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{8} = \frac{\pi}{4} = 45^\circ$ $\therefore E_i = E_0 \angle 45^\circ$

Ily at $s = \frac{\lambda}{4}$; $E_i = E_0 \angle 90^\circ$

Vector diagram for incident wave



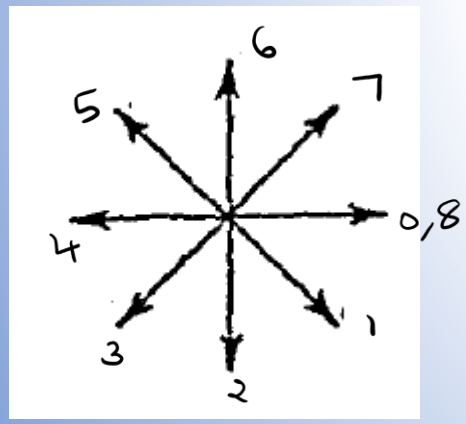
Linear diagram for incident wave (t=0)



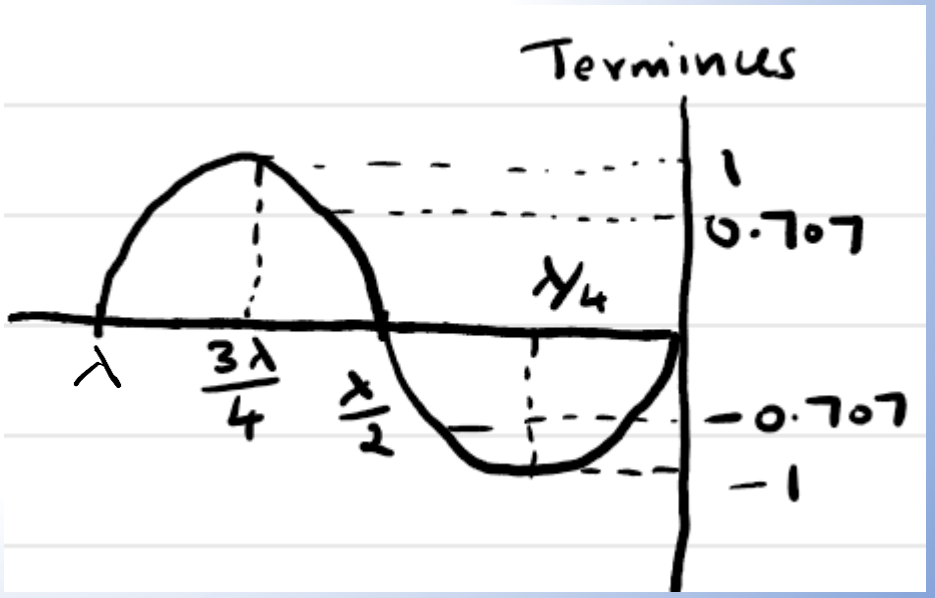
- Reflected wave $E_r = E_0 e^{-j\beta s}. e^{j\omega t}$
- For K = 1 (open circuited line)
- Er = Eo $\angle \omega t - \beta s$

For $t=0$, $E_r = E_0 \angle -\beta s$

Vector diagram for reflected wave



Linear diagram for reflected wave (t=0)



At t = π/8

For incident wave

$E_i = E_0 \angle \beta s + 45^\circ$

At $s = 0$, $E_i = E_0 \angle 45^\circ$

$s = \frac{\lambda}{8}$, $E_i = E_0 \angle 90^\circ$

$s = \frac{\lambda}{4}$, $E_i = E_0 \angle 135^\circ$

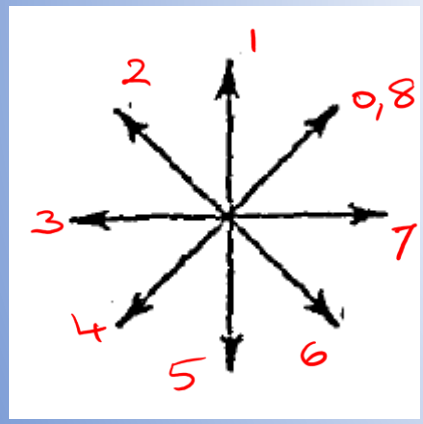
For reflected wave

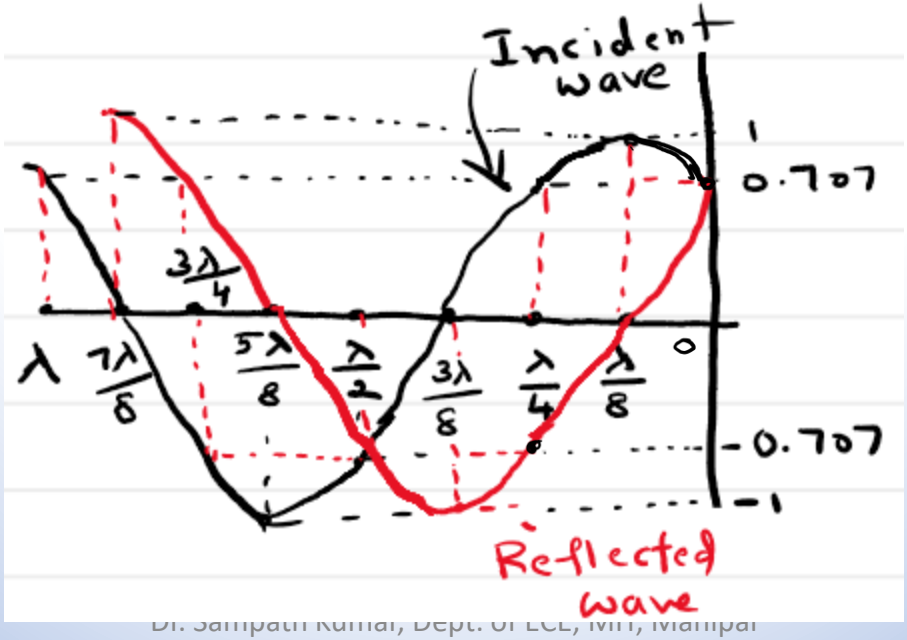
$E_r = E_0 \angle -\beta s + 45^\circ$

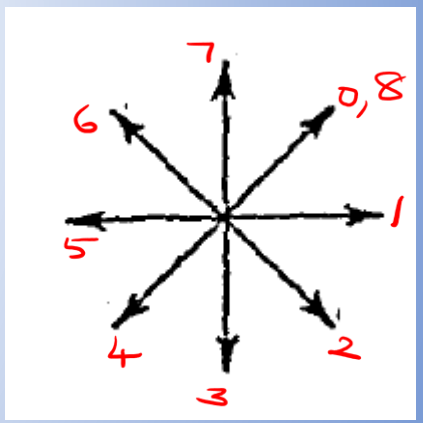
At $s = 0$, $E_r = E_0 \angle 45^\circ$

$s = \frac{\lambda}{8}$, $E_r = E_0 \angle 0^\circ$

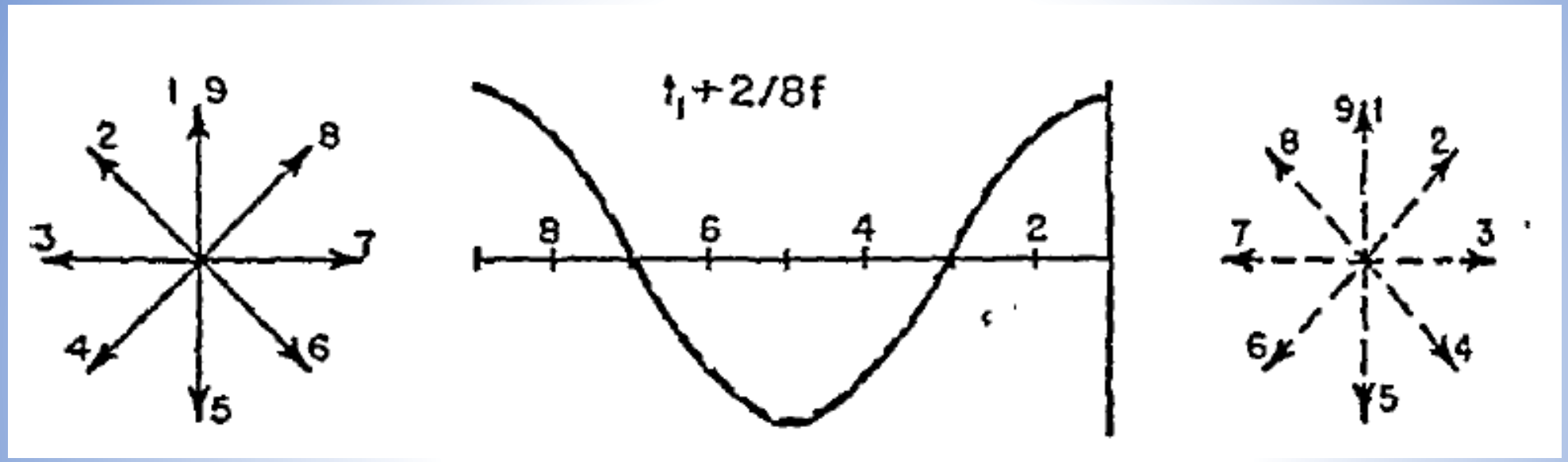
$s = \frac{\lambda}{4}$, $E_r = E_0 \angle -45^\circ$

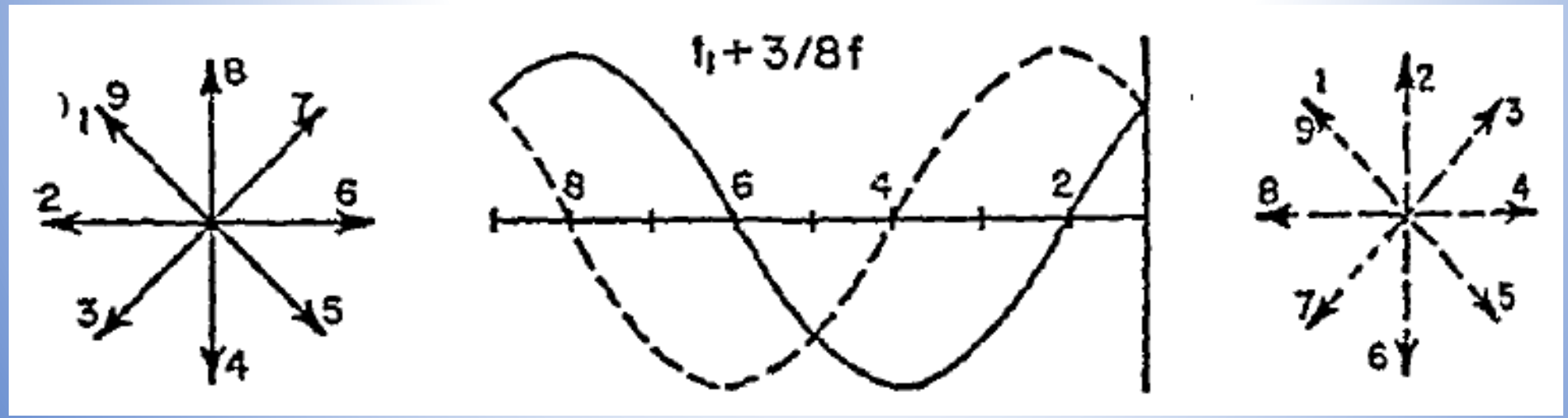




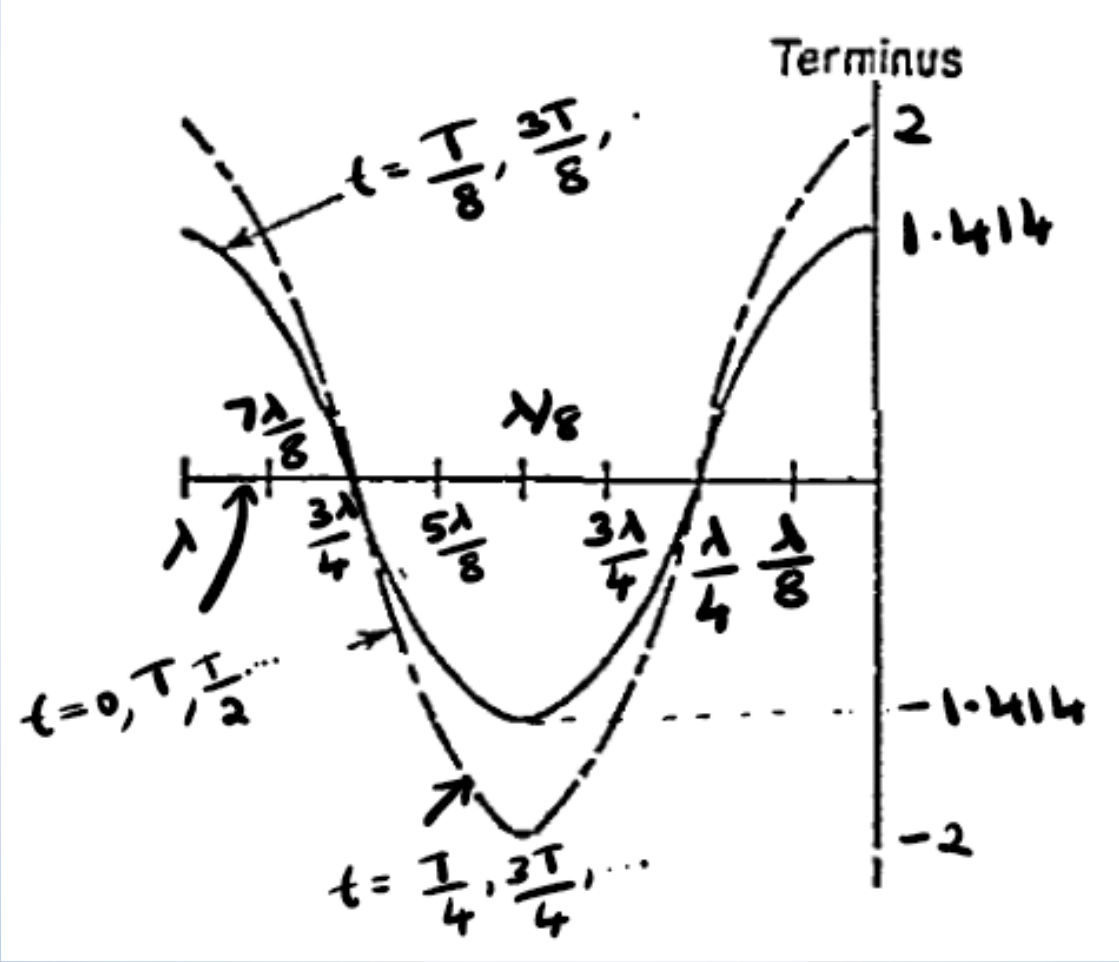


Draw at t = 2π/8 and 3π/8

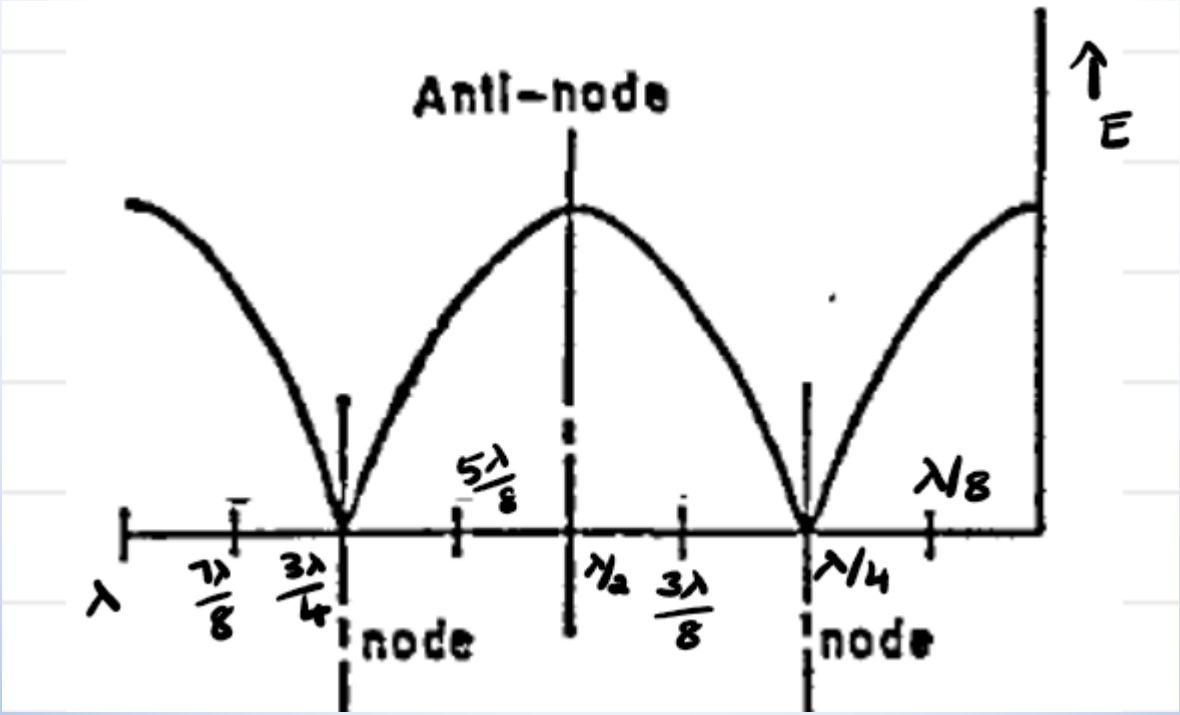




- The actual voltage at any point on the TL is the sum of incident and reflected wave voltages at that point



- If the resultant total voltage appears to be standstill, oscillating with in magnitude with time but having fixed positions of maxima and minima
- Such a wave is known as a standing wave
- If the line voltages are measured with an RMS voltmeter, the magnitude will appear as shown below.
- Note: Voltmeter doesn't distinguish between +ve and –ve values




Thank
you